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# Tariffs as Environmental Protection: Evidence from the Global South after the China Garbage Shock

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## Abstract

In the global waste trade, importers buy containers of waste and scrap to meet demand for raw materials, especially in the Global South. But post-processing leftovers generate localized negative externalities. I use the waste trade as a setting to establish that low-capacity states can and do use tariffs as a tool in their environmental policy repertoire. Product-level tariffs can serve as Pigouvian ‘sin’ taxes that incentivize private market actors to limit transactions and/or increase state revenue, both channels that can result in improved environmental outcomes. For evidence, I leverage the ‘China garbage shock’: in 2017 China banned imports of twenty-six waste products (HS six-digit), which disrupted economic–environmental trade-offs in other, newly competitive markets awash in diverted imports. Using novel data on 179 traded waste products and product-level tariffs (1996–2020), I demonstrate that those that received the shock raised tariffs in ways consistent with environmental protection.

**Keywords:** tariff; environment; waste trade; Global South; China

## Introduction

In the global waste trade, exporters collect post-consumer materials and post-industrial byproducts and offer them for sale on international markets. Importers buy containers of waste products, because they can process the waste and scrap to recover valuable raw materials. While raw materials are sourced from waste worldwide, resource-scarce developing states have proven especially eager for imported waste products, and developed states have leaned into the ‘not in my backyard’ (NIMBY) benefits to be had from designing waste management around export. Based on a list of 179 internationally traded waste products introduced in this article, the waste trade has netted out to over 1 billion tons of waste products exported from Global North states and imported into Global South states since 1995.<sup>1</sup>

For decades, China has bought waste products en masse. As recently as April 2017, a prominent executive in the globalized waste and recycling industry explained that ‘the industry is dependent on China for moving of materials’. So the world was turned upside down when, on 18 July 2017, China banned imports of twenty-six specific waste products as part of its ‘Operation National Sword’ – also known as ‘No More Foreign Garbage’. Import bans on unsorted waste paper and post-consumer plastics would be fully implemented in the five months before the end of 2017; the

<sup>1</sup>‘Global North’ states are, as of 1995, (1) OECD members and (2) classified as High Income by the World Bank. Appendix A.

implementation timeline for other bans on waste products of metal, animal, and textile origin was delegated to regulators, although the goal of speed was implied. Chinese imports of banned waste products plummeted.<sup>2</sup> A large literature considers the political-economic consequences of China's accession to the World Trade Organization (WTO) in 2001 and, more generally, China's entry into global markets. This has been called the 'China shock' (Autor et al. 2016). In contrast, the 'China garbage shock' of 2017 affords the opportunity to see political-economic consequences when China exits.

With Chinese buyers regulated out of business, importers of China ban products in other Global South states experienced a positive shock to their competitiveness. The China garbage shock was thus a boon for other Global South states' tradeable sectors engaged in waste product import and processing, not to mention their domestic downstream users of recovered raw materials. However, importers' private transactions also relocated the valueless dirt and debris that arrives in a container of waste products. This content is known as 'end-of-life' (EOL) waste, which must be disentangled before recovered raw materials can be used as inputs in further production processes. The China garbage shock meant that unprecedented amounts of foreign-origin EOL waste were now going to spend their afterlife in these states' domestic waste management systems.

Low-capacity states struggle to implement regulations capable of mitigating domestically generated environmental harms, much less harms that result from economic integration (Dolšák and Prakash 2022; Rudra et al. 2018; Prakash and Potoski 2014; Boudier and Bensebaa 2011). Waste management is a particular burden: according to the World Bank, over two-thirds of Global South garbage is disposed of in open dumps. In low-income states, over 90 per cent is in open dumps. For comparison, only 2 per cent of waste in high-income states is in open dumps (Kaza et al. 2018). That massive amounts of EOL waste are of foreign origin is obvious to people throughout the Global South. Imported dreck chokes Southeast Asian rivers, factories burn imported plastic for fuel at temperatures far too low to avoid toxic fumes, untrained workers dismantling cruise ships are killed and maimed, and discarded clothes in Chile's Atacama desert are now visible from space.

Global South states on the receiving end of the China garbage shock thus found themselves at a crossroads: this tradeable sector boomed, but so too did its undeniable domestic environmental harms. It is a truism in political economy that states must balance the interests of domestic tradeable sectors against the interests of those threatened by trade and thus seeking protection, whether or not that includes the environment. In fact, there is a classic trade policy lever available to governments navigating these trade-offs: the product-level tariff. I use the waste trade as a setting to demonstrate that tariffs can generate environmental protection, too.

Conceptually, a tariff can serve as a Pigouvian 'sin' tax that just happens to be levied at a border (Pigou 1932; Wiseman and Ellig 2007). 'Sin tariffs' can generate direct and indirect environmental benefits. Directly, a higher sin tariff makes it more expensive to import a polluting product, which improves local environmental protection by discouraging imports by those who now find it too expensive. Fewer waste product imports means less pollution manifests in the would-be importer's jurisdiction. Indirectly, a sin tariff extracts more compensation from those actors deriving private benefits from continued import of polluting products. Growing state coffers have the potential to improve state capacity which can have knock-on environmental benefits, whether or not funds from sin tariffs are actually earmarked for and spent on mitigating pollution. Via this combination of direct and indirect channels, sin tariffs can generate environmental protection in the government's territory. This holds true even if the government has no sincere interest in environmental protection whatsoever.

The waste trade is an especially useful setting in which to uncover sin tariffs, as the concept of domestic import-competers – the theoretical backbone explaining protectionist trade policy – is out of place. The global waste trade has arisen as raw material needs outstrip domestic supply,

<sup>2</sup>Appendix D. Staub, Colin. 25 July 2017. 'How WM and other exporters are reacting to China's ban'. Resource Recycling.

whether from virgin or secondary sources (Minter 2015; O'Neill 2019; Goldstein 2020). At the same time, the Global South response to the China garbage shock provides a most-likely case of observing sin tariffs in operation, as the shock provides an impetus to adjust trade policy given that both domestic rents and domestic environmental harm have multiplied relative to the pre-shock status quo. I use a difference-in-differences (DiD) research design to demonstrate that Global South state–waste product combinations ‘treated’ by the 2017 China garbage shock set higher tariffs compared to those left ‘untreated’. In doing so, I use novel data on 179 internationally traded waste products, nearly triple previously best-available catalogs of waste products, and I match these waste products to newly assembled data on product-level import tariffs for 170 Global South states (1996–2020). This approach has the further advantage of revealing evidence of sin tariffs at scale – not just in any particular policy experiment by any particular state, but in the data across time and space. For contemporary qualitative evidence that actors in ‘treated’ states understood the economic–environmental trade-offs embedded in sin tariffs, I draw on Malaysia, a key state on the receiving end of the China garbage shock.

To emphasize, I argue that a state’s sin tariff can further environmental protection *in that state’s territory*. That Global South states can redistribute negative externalities outside of their borders, even using an old-fashioned trade policy instrument, is an important finding and one that speaks to their power within a capitalist international political economy. At the same time, the effects of protectionism by individual Global South states on global environmental outcomes are at best ambiguous. A waste product has to go somewhere once produced, and the production of waste products is, in a word, inelastic. Might Global South restrictions have wider effects, even on those generating waste products in the Global North (such as the author)? I engage with this and other open questions in rubbish conclusions.

### Sin Tariffs and the Waste Trade

A Pigouvian ‘sin’ tax is a form of taxation used to reduce the negative externalities caused by socially and thus politically undesirable activities (Pigou 1932; Banzhaf 2020). A higher tax increases the cost of engaging in the activity, with the dual effects of reducing the production of negative externalities should private actors now find the activity too expensive, and extracting more compensation from those that still engage in the activity and thus continue to derive private benefits from it. A sin tax can be applied within a jurisdiction or at a border between jurisdictions (Wiseman and Ellig 2007). Think of a ‘sin tariff’ as an import tariff that increases the costs to bringing a negative externality-laden good across an international border. A higher sin tariff can reduce the production of domestic negative externalities by making the offending product too costly to import, relative to the counterfactual in which domestic negative externalities are produced by virtue of importing the container. And/or, a higher sin tariff can extract more revenue for government coffers from those commercial actors that continue to derive private benefits from trading the offending product. Such increased tariff revenue can indirectly mitigate negative externalities, for example by improving state capacity or furthering economic development, even if not earmarked for specific trade-related harm mitigation. Perhaps most importantly, the sincerity of the government’s commitment to mitigating ‘sin’ is not at issue. Extracting the tax in itself can have mitigating effects insofar as it changes the incentives faced by those engaged in the trade (Pigou 1932).

A sin tariff has a variety of desirable properties that make it an attractive tool to mitigate the domestic manifestation of foreign-origin negative externalities. Because the tariff is a unilateral action, positive environmental effects can accrue whatever the status of interstate co-ordination and global governance. Tariffs are incremental, so a government can set a tariff low enough to accommodate rather than price out domestic winners from trade. They are easily adjustable, so the government can change its policy in response to changing political and economic conditions.

They are also targeted, so the government can make trade-offs on a product-by-product basis (Kim et al. 2020). Not to be forgotten, a tariff can be useful politically as a tool that at least *appears* to attach costs to foreign actors, whatever a post hoc tallying of tariff pass-through might reveal (Mansfield and Solodoch 2024). I return to the question of tariff pass-through in the conclusion.

Product-level tariffs slot into a standard customs infrastructure, whereas domestic environmental policy alternatives can be costly, difficult to implement, and subject to manipulation (Rudra 2008; Perlman 2020; Gulotty 2020). Think for example of the resources needed to have inspectors open containers, test whether the qualities and percentages of contaminants therein meet regulatory thresholds, and impose penalties for violations. Another policy alternative would be to enforce health and safety regulations ‘behind-the-border’ at importers’ processing facilities. Doing so would not only require significant regulatory capacity, but it would also forego the possibility of addressing foreign-origin pollution separately, when it is still packed in a container at the port, and before what was once foreign becomes physically and politically co-mingled with domestic waste and its management.

For a sin tariff to be useful, a state must be constrained by a trade-off between economic and socio-environmental goals in making its decision over openness to a given product. One can imagine many potentially ‘sinful’ imports: flora and fauna that could become invasive species; weapons with the potential to generate harms for natural and social environments; single-use goods destined for landfills for which there is no secondary demand for recovered raw materials; or goods carrying negative externalities for climate. I hone in on legally traded waste products as exemplars of this class of imports. In the Global South especially, openness to trade in waste products trades off undeniable domestic demand against real environmental harms from limited domestic waste management capacity (Kaza et al. 2018). Further, the negative externalities of imported EOL waste are not inherently transboundary; rather, they are highly localized, often manifesting via literal piles. A sin tariff targeting EOL waste can mitigate the domestic production of socio-environmental harms, irrespective of any impact it might have on environmental outcomes outside the state’s territory.

On the economic side, the waste trade has a potent set of winners in Global South domestic political economies. The actual importers signing contracts and taking possession of containers of waste products are just one set of winners. Securing domestic access via imported waste products is valuable in geographies scarce in raw materials, and deriving raw materials from waste products is not inferior in itself (Gregson and Crang 2015; Xu et al. 2022). If recovered raw materials are deployed as inputs to exports, the waste trade carries further benefits for accruing foreign exchange. Further, an estimated 20 million ‘waste pickers’ worldwide do the labor-intensive work of sorting and processing waste products and thus have a direct, material stake in the continued import of waste products.<sup>3</sup> Many if not most waste pickers are informal, extremely poor, and politically marginalized.<sup>4</sup> Nonetheless, waste pickers in the Global South are archetypes of the Heckscher–Ohlin model of trade preferences: low-skilled workers in states abundant in that factor are predicted to be strongly pro-trade, as the kinds of low-skill-intensive goods they produce are competitive on world markets. Consistent with this expectation, transnational social movements representing waste pickers argue for a ‘just transition’ focusing attention on formalizing employment if and when global regulation of the waste trade increases.<sup>5</sup>

Several other characteristics of the waste trade also make it a useful setting to observe sin tariffs. First, research on the political economy of trade points us to domestic import-competing interests as a key determinant of trade protection (Kim 2017; Kim et al. 2020). However, as waste products

<sup>3</sup>Bureau of International Recycling (2023).

<sup>4</sup>To date, there is only exceptional evidence that waste picker collective action has successfully altered domestic policy (Gutberlet et al. 2017; Amuzu 2018).

<sup>5</sup>See, for example, statements by the International Alliance of Waste Pickers at UN Environment Assembly 5.2, November 2023 (<https://globalrec.org/global-meeting/unea5-kenya/>).

are byproducts, import competition is reasonably a second-order worry for domestic producers in direct competition with imports. Import competition might matter for domestic producers of virgin raw materials. Still, international markets for waste products have arisen as raw material needs outstrip domestic supplies of raw materials, from whatever source (Minter 2015; O'Neill 2019; Goldstein 2020). We can reasonably infer that what tariffs exist on waste imports require more explanation than a story of direct material losers worried about being out-competed.

Second, global governance around the waste trade is limited. The most successful effort is the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, with the core function of providing repositories for member state self-reporting on waste shipments. However, while the Basel Convention was signed in 1989 and quickly reached near-universal coverage, the United States is not a member.<sup>6</sup> The Basel Convention facilitates a program of technical work with regard to 'toxic, poisonous, explosive, corrosive, flammable, ecotoxic, and infectious wastes', but radioactive waste is excluded. Although twenty-seven African states are party to the 1998 Bamako Convention that bans hazardous (including radioactive) waste imports, there are serious concerns over compliance (Okafor-Yarwood and Adewumi 2020). The UN began negotiating a binding treaty on plastics in 2022, but it failed to conclude negotiations in 2024 as planned; one sticking point is the extent to which the treaty's goal is to reduce plastic production or increase plastic recycling. In short, Global South states looking to curb the waste trade's excesses have reason to turn to domestic alternatives.

Indeed, contemporary Global South states have passed over 1,200 domestic non-tariff measures (NTMs) that increase regulation of the waste trade (1996–2020).<sup>7</sup> Yet a third characteristic of the waste trade – that it is a pipeline to access raw materials – suggests it is especially difficult for domestic environmental laws to bite in the Global South. The 'Environmental Kuznets Curve' (EKC) posits that richer states past some threshold can afford to forego economic activity in favor of environmental protection, but poorer states are compelled to accept pollution as a correlate of development. EKC mechanisms and causality are in question (Stern 2004), though when plotted against GDP per capita waste imports do exhibit the telltale inverted U-shape relationship.<sup>8</sup> EKC constraints on regulatory appetites are likely especially germane around the waste trade: raw materials derived from waste products become direct inputs into construction, industrialization, manufacturing, and other facets of economic development. Despite such limitations, the popularity of waste trade NTMs suggests that Global South states consider these important international environmental policy tools, alongside ongoing global governance efforts. What I point to is an adjustable, incremental, targeted, unilateral, easy-to-implement, potentially revenue-generating option that might be old-fashioned but should not be overlooked.

## Garbage Data

What are these waste products that are bought and sold around the world? Table 1 lists the waste products subject to the China garbage shock, which include several ash, residue, and slag products; several yarn and textile waste products made of cotton, wool and animal hair, or artificial fibers; sorted and unsorted rags; unsorted paper; and all post-consumer plastics products. China defined the waste products in Table 1 with Harmonized System (HS) codes, which is the statistical infrastructure maintained by the World Customs Organization's (WCO) Harmonized Commodity Description and Coding System. HS codes are hierarchical and move from

<sup>6</sup>New hazardous plastics amendments that came into force in 2019 carry promise for increasing Basel's impact, though see Yang (2020). Membership in the subsequent 1998 Rotterdam Convention on pesticides and industrial chemicals and the 2001 Stockholm Convention on persistent organic pollutants is in the dozens.

<sup>7</sup>Appendix F. Data collected from UN FAOLEX.

<sup>8</sup>Appendix G.

**Table 1.** Waste product imports banned in China’s ‘Operation National Sword’ (2017)

| Type    | Description                                         | HS codes       |
|---------|-----------------------------------------------------|----------------|
| Animal  | Wool and fine animal hair waste                     | 510310; 510320 |
| Animal  | Coarse animal hair waste                            | 510330         |
| Animal  | Fine and coarse animal hair waste (garnetted stock) | 510400         |
| Metal   | Slag and waste from iron and steel manufacturing    | 261900         |
| Metal   | Ash and residue containing mainly zinc              | 262011; 262019 |
| Metal   | Leaded gasoline sludges                             | 262021         |
| Metal   | Lead residue (other than leaded gasoline sludges)   | 262029         |
| Metal   | Ash and residue containing mainly copper            | 262030         |
| Metal   | Ash and residue containing mainly aluminum          | 262040         |
| Metal   | Mixed residues and ash, non-ferrous including toxic | 262060         |
| Metal   | Mixed residues and ash, non-ferrous and non-toxic   | 262091; 262099 |
| Paper   | Unsorted paper waste and scrap                      | 470790         |
| Plastic | Ethylene polymer waste, pairings, and scrap         | 391510         |
| Plastic | Styrene polymer waste, pairings, and scrap          | 391520         |
| Plastic | Vinyl chloride waste, pairings, and scrap           | 391530         |
| Plastic | Plastic waste and scrap ‘not elsewhere specified’   | 391590         |
| Textile | Cotton waste (yarn and thread)                      | 520210         |
| Textile | Cotton waste (garnetted stock)                      | 520291         |
| Textile | Cotton waste, ‘other than ...’                      | 520299         |
| Textile | Synthetic fiber waste (unspecified)                 | 550510; 550520 |
| Textile | Rags, sorted                                        | 631010         |
| Textile | Rags, unsorted                                      | 631090         |

Note: Chinese notification to the WTO, 18 July 2017 (G/TBT/N/CHN/1211).

categories (two- and four-digit) to a specific product with a six-digit HS code.<sup>9</sup> In a response to WTO complaints over its actions, China explained, ‘Currently, there is not any globally recognized standard for scrap materials and recyclable materials. China therefore adopts the universally applicable HS code’.<sup>10</sup>

China is correct: unfortunately, especially in light of its normative salience, the global waste trade is poorly measured (O’Neill 2019). Thus, it requires a measurement innovation to understand the China import bans in the context of all legally traded waste products. I introduce a list of 179 traded waste products also at the HS six-digit level, documented in HS revisions through 2017, which nearly triples the previously best-available list in the OECD Trade in Waste and Scrap database.<sup>11</sup> To be classified as a waste product, the definition includes the term *waste* or *scrap*;<sup>12</sup> it is a residual or byproduct from primary production processes;<sup>13</sup> and/or the product is a one-time primary good intended to be processed into inputs for further use.<sup>14</sup> Consistent with industry practice, waste products are categorized into types, based on their origin: animal, chemical, metal, mineral, paper, plastic, textile, or vegetable. Appendix B provides all included HS codes and further examples for the interested reader.

Some waste products are perhaps eyebrow-raising, such as trade in sewage sludge (HS 382520). Why buy that? An importer can acquire foreign-origin sewage, which may be better treated or

<sup>9</sup>Codes beyond six digits are not globally harmonized. For more on the WCO and HS codes, see Betz (2019).

<sup>10</sup>Staub, Colin. 19 June 2018. ‘China: Waste restrictions will spur US job growth’. Resource Recycling.

<sup>11</sup>The OECD list of sixty-three misses two-thirds of products China banned in 2017 (Kellenberg 2012). HS codes have been revised every four to five years. These data cover revisions applicable in the study period (1996–2020). The maximum number of waste products in any one revision is 144 (2002 revision), and the minimum is 129 (1992 revision).

<sup>12</sup>For example, HS 470730: Paper or paperboard: waste and scrap, of paper or paperboard made mainly of mechanical pulp (e.g. newspapers, journals, and similar printed matter).

<sup>13</sup>For example, HS 382530: Residual products of the chemical or allied industries, not elsewhere specified or included: clinical waste.

<sup>14</sup>For example, HS 401220: Rubber: used pneumatic tires. In contrast, used goods that are directly intended for resale – such as used vehicles in working order – are not waste products.



have more consistent qualities than domestic sewage. The importer can then perform a value-added service to process it into fertilizer. Such on-shoring of labor-intensive and low-technology services can be more profitable to the importer than buying fertilizer itself, not to mention attractive to an exporter in a high-cost jurisdiction. We can expect the exporter and importer to bargain over the price of the waste product, taking this service provision into account. In terms of externalities, trade in sewage is a good example of the ‘not in my backyard’ (NIMBY) benefits of getting post-processing leftovers out of the exporter’s jurisdiction, and ‘yes in my backyard’ (YIMBY) costs of accepting them. What sewage has in common with all other traded waste products is that the importer knowingly, intentionally buys EOL content mixed in with the raw materials of value. Because sewage has an HS code like the other waste products in the dataset, we can track international movements via exports and imports – and, as a consequence, the global redistribution of EOL components.

To be clear, whether an international regulatory body like the WCO provides an HS code for something and therefore is capable of monitoring it is endogenous to politics (Buthe and Mattli 2014). Of special importance, e-waste products only received their first dedicated HS codes in 2022 (outside the study period), meaning that what we know about the movement of e-waste around the world has been incredibly limited (Lepawsky 2018).<sup>15</sup> The multiple oceanic Garbage Patches illustrate that plastic waste has been notoriously poorly accounted for (Lebreton et al. 2018). Additionally, HS codes do not account for trade in services, so the list of waste products does not include waste exported at a negative price (in which the service of foreign waste disposal is imported). Illegal waste dumping – trade in services at a too-low price – is also outside the scope. Still, it is crucial to advance the research agenda on the political economy of the global waste trade now, starting with the data that states have endogenously chosen to collect.

To probe the sin tariff theory, I compile the product–state–year applied most favored nation (MFN) import tariff for each of these 179 waste products (HS six-digit), for up to 170 Global South states.<sup>16</sup> To provide insight into the data, Figure 1 presents heat maps that capture the maximum and minimum tariff for each Global South state–waste product combination during the study period (1996–2020). States are organized alphabetically within region on the y-axes, and waste product HS codes are organized sequentially within type on the x-axes. Colors correspond to a log scale in which the lightest color is equivalent to a 0 per cent tariff and the darkest color indicates a 100 per cent or greater tariff.<sup>17</sup> A first takeaway from Figure 1 is that, as implied by the sin tariff theory, Global South states are indeed setting lots of line-item tariffs on waste products. Specifically, there is at least one tariff value reported in 71 per cent of cells. Unfortunately, as tariff data ultimately rely on state reporting, we cannot adjudicate what proportion of missingness is due to the true absence of a product-level tariff (Barari and Kim 2022). However, we can benchmark against the share of tariff lines with specific rates across all traded products. In fact, among the state-years in the sample here, the median tariff line coverage across all products is less than 1 per cent, and even at the 99th percentile coverage is only 35 per cent.<sup>18</sup>

Second, patterns in Figure 1 are consistent with various observable implications that should hold if and when sin tariffs are an attractive and useful policy tool in the waste trade. Comparing across rows in Figure 1, we see that Global South states’ tariff policies towards waste products are

<sup>15</sup>The World Customs Organization acknowledged the e-waste measurement problem, especially as e-waste is an ‘example of a product class which presents significant policy concerns as well as a high value of trade’. HS Nomenclature 2022 edition, Section XVI, New Note 6, (A)(ii). UN and other intergovernmental bodies are collaborating on the Global E-Waste Statistics Partnership to improve measurement and awareness ([globalewaste.org](http://globalewaste.org)).

<sup>16</sup>Average WITS (TRAINS) ad valorem (or equivalent) duties expressed in percentage terms averaged across all directed dyads for a given product–state–year. Enormous thanks to In Song Kim and collaborators for organizing these data (Barari and Kim 2022).

<sup>17</sup>In 36 per cent of product–state–year observations the tariff is zero, meaning that the state set a tariff and chose that tariff to be 0 per cent (as opposed to not setting a product-level tariff). In 402 instances, the tariff is higher than 100 per cent.

<sup>18</sup>World Development Indicators, TM.TAX.MRCH.SR.ZS.

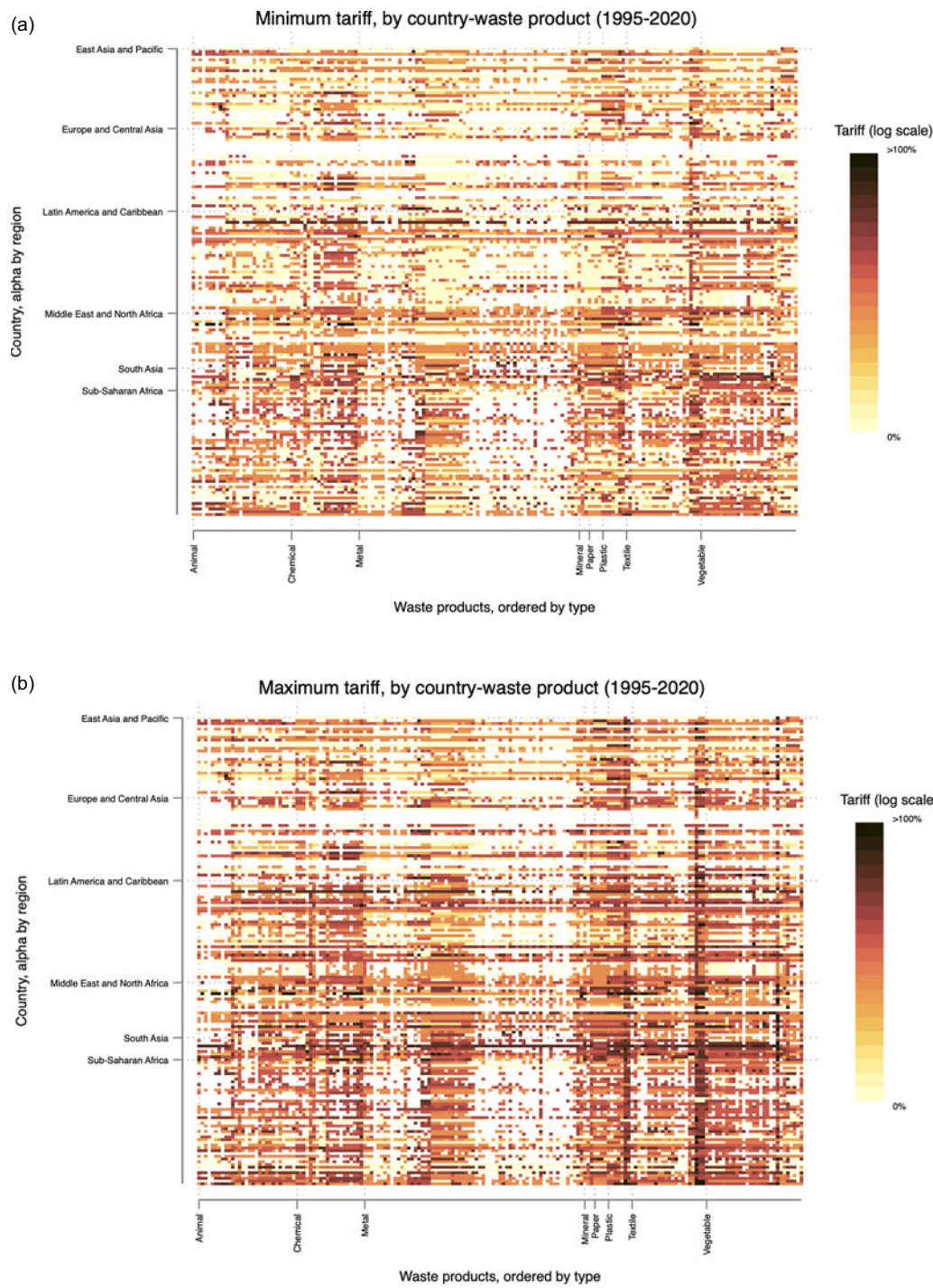


Figure 1. Heat maps demonstrating variation in (a) minimum and (b) maximum waste product tariffs (1996–2020).



differentiated, reflecting that tariffs are unilateral policy actions that require no co-ordination. Looking within rows, we see that effectively no state has one tariff policy towards waste products, but rather differentiates product-level tariffs as implied by the sin tariff theory. When comparing the same cell in Figures 1(a) and 1(b), we see evidence that states are not just setting and forgetting tariffs on waste products, but are adjusting them to changing conditions, as they should if and when the tariff were a useful tool for adapting to changing economic–environmental trade-offs. Looking down the columns in Figure 1, we see that states’ propensity to set tariffs at different levels varies by waste type. These patterns reflect the reality that types vary in many ways relevant to choices over taxation: processing them requires different combinations of labor and capital; the availability of virgin raw material alternatives differ; and the qualities of and thus harms generated by EOL components differ as well. Still, the significant variation within waste types reinforces that political choices over tariffs are about more than the physical substance in question.

Figure 2 aggregates the granular tariff data to summarize over-time trends, and for comparison it includes the smoothed average tariff for Global South states in the sample. It is well-known that tariff levels have been trending downward; for this Global South sample and period, the average declined from around 10 per cent to around 5 per cent. Descriptively, waste product tariffs have also declined in the period, although not strictly so and at different rates across types. For more descriptive insight into the tariff data, see Appendix E. Understanding effects of the fallout of the 2017 China garbage shock on trends in product-level tariffs is the subject of the next section.

### China Garbage Shock

The precise timing of China’s ‘Operation National Sword’ waste import bans was a product of issue linkage, serving as one of China’s opening salvos in the United States–China trade war. In response to US criticism, a Chinese spokesperson said, ‘The US officials are so hypocritical as they try to make an issue out of China’s legitimate and lawful actions and accuse China of “seemingly violating WTO obligations” . . . How could they claim that their restrictions on exports of high-tech and high value-added products to China are legitimate while China’s lawful restrictions on its imports of foreign garbage are illegal?’<sup>19</sup> Domestically, China paired the import bans with a massive reorganization of its internal recycling networks to formalize the sector and improve indigenous waste utilization (Goldstein 2020, 234–36). While domestic reforms might have been foreseeable, not just the timing but also the substance of China’s international trade actions was surprising. The global waste industry thought that China’s tightening of enforcement four years earlier in ‘Operation Green Fence’ had been successful enough to preclude further trade-related regulation anytime soon, and certainly not in the form of outright import bans. Even *ex post* rationales for why these specific HS codes were targeted are absent from industry discourse. Further, there is anecdotal evidence that even Chinese firms did not fully predict the specific targets of the bans. A publicly acknowledged meeting between Chinese industry representatives and the Chinese state took place in April 2017; reporting at the time suggests that the Chinese waste paper industry was wary of possible future import quotas but not bans, whereas waste plastic importers were sanguine. Only in the aftermath did Chinese paper and plastics recycling firms expand into foreign plants, receiving and processing waste imports abroad so that the products they then shipped into China via intra-firm trade were no longer waste.<sup>20</sup>

While there is much to explore about the consequences of the China garbage shock, I focus on using it to support observable implications of a theory of sin tariffs: Global South states for which China’s 2017 action generated a simultaneous positive shock to competitiveness and negative shock to the environment should turn to higher tariffs, all else equal. The situation lends itself to a

<sup>19</sup>Staub, Colin. 27 March 2018. ‘China bites back after diplomats criticize scrap ban’. Resource Recycling.

<sup>20</sup>Clapp (2025); Goldstein (2020), ch 8. See also: Staub, Colin. 3 April 2018. ‘Policies drive Chinese processing investments across the globe’. Resource Recycling.

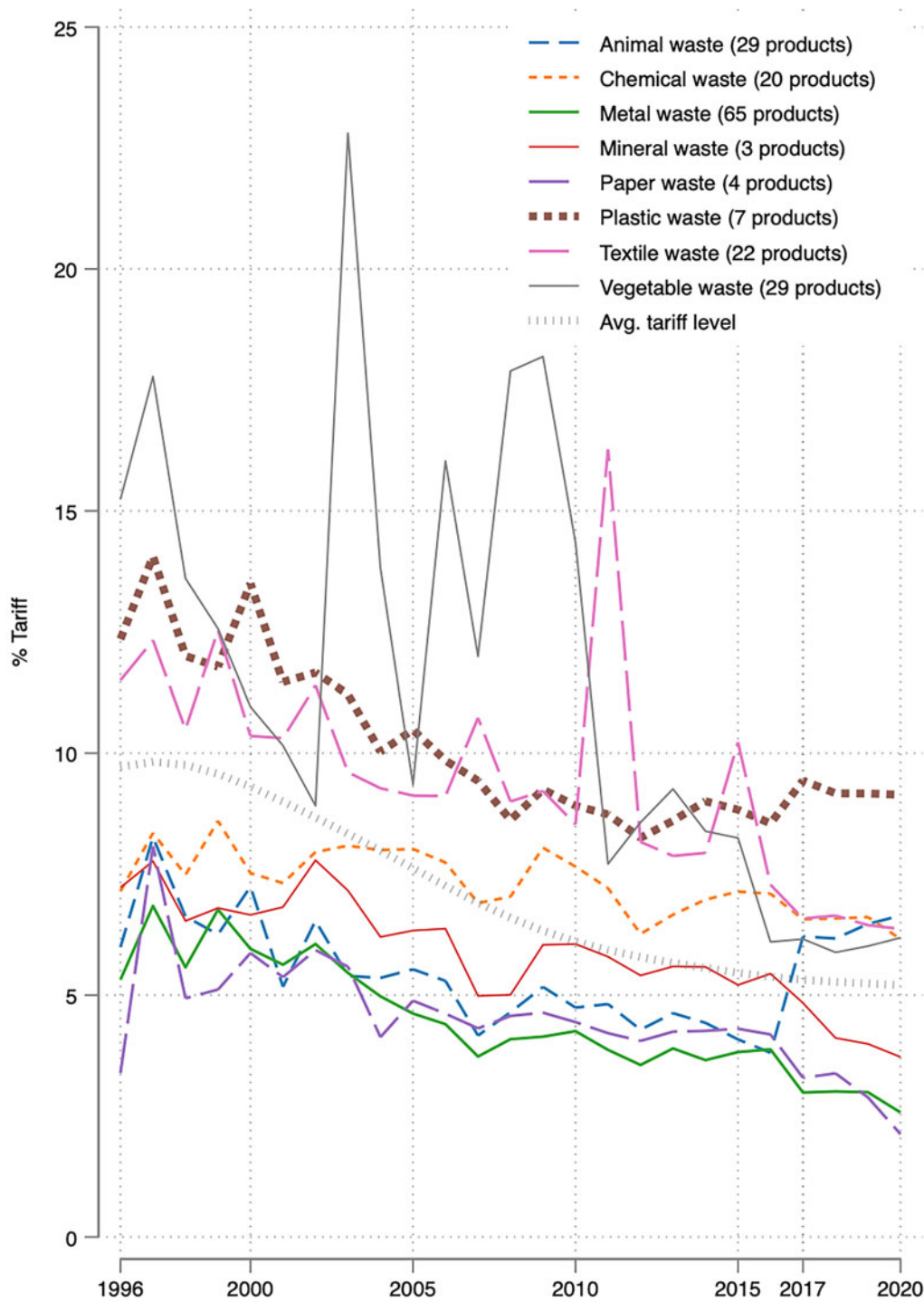


Figure 2. Waste product tariffs trends, by waste type-year (1996–2020).

difference-in-differences (DiD) estimation of the average treatment effect on the treated (ATET), or the effect on the tariff of developing state-banned product combinations ‘treated’ with trade diversion, compared to the counterfactual in which those developing state-banned product combinations were not treated. For estimates to be interpreted causally, we must be convinced that the shock was exogenous. For DiD to be identified, we must be satisfied with the ‘parallel trends’ assumption: the trends in tariffs for treated and untreated developing state-banned product combinations were parallel before the China garbage shock, and they would have remained parallel if not for the China garbage shock.

I conceptualize the treatment geographically: in commercial markets, exporters were likely to divert banned waste products to importers in China’s neighborhood, especially in an emergency situation when ships at sea suddenly needed new ports of entry. Indeed, a number of careful studies have done the work of tracing trade diversion for banned waste products, confirming that they especially flooded neighboring states (Ma et al. 2021; Pacini et al. 2021; Tran et al. 2021; Brooks et al. 2018; Velut and Wellhausen 2026). Therefore, I apply the treatment to combinations of Asia-Pacific developing states and banned waste products. Is it credible to understand the bans as exogenous from the point of view of Asia-Pacific states? Of all states in the world, surely China’s neighbors are motivated to anticipate a shock that would divert foreigners’ garbage from China to their own shores. However, the trade-war timing of the bans was likely unpredictable enough to weaken the ability of even China’s neighbors to anticipate the action. Additionally, the geographic conceptualization of the treatment marks as treated many developing state-banned product combinations that were in fact unlikely destinations for diverted imports, introducing attenuation bias.

We might be concerned that there is heterogeneity in the sample of 179 waste products that complicates the counterfactual. Even if the twenty-six targeted products were plausibly unknown *ex ante*, it may be generally understood that imports of some of the 179 waste products in the sample would never be banned. On this issue, China offers a research design lifeline: after the initial China garbage shock, China announced its intention to ban imports of an additional set of eighteen waste products either at the end of 2018 or 2019. However, China did not follow through with these bans in the study period (or to date). Interestingly, these announcements were not widely reported; real-time industry analysts cite English translations and HS codes that are marked as ‘for reference only’.<sup>21</sup> Nonetheless, I take the announcements as credible indications that, in a counterfactual world, China might have banned imports of these additional waste products, too. This subsample is thus especially compelling in establishing that policy adjustment given exposure to trade diversion is the mechanism at play.

On parallel trends, Figure 2 shows the secular downward trend in the study period, with average tariffs for some but not all waste types trending similarly. By implication, the China garbage shock may be causing absolute increases due to the treatment and/or smaller downward adjustments for treated versus untreated combinations. In general, tariffs are complicated outcomes, and huge interdisciplinary literatures speak to their determinants. It is reasonable to expect that confounders may pollute the assumption of parallel trends (see Appendix H for visualizations). Therefore, I estimate models both without and with a slate of covariates and fixed effects. These include a product-level control for the size of the product’s world market (USD value of exports + imports, ln). At the state level, I control for factors known to influence propensities towards trade liberalization: regime type, trade per GDP (ln), WTO membership, and unemployment (ln). I also control for GDP per capita and its square, per the ‘Environmental Kuznets Curve’ (EKC) proposition. Several of these controls also account for heterogeneity in endowments of recycling technologies and technical capacity in importing states. I add controls

<sup>21</sup>Appendix C. Unlike the HS six-digit waste products notified to the WTO in 2017, the announcement lists waste products at higher levels of disaggregation (HS ten-digit). If and when the overarching HS six-digit codes include subcategories that were never intended to be banned, attenuation bias is introduced.

**Table 2.** Higher tariffs on imports treated by China garbage shock

|                                | (1)                  | (2)                    | (3)                  | (4)                    |
|--------------------------------|----------------------|------------------------|----------------------|------------------------|
|                                | All products         | Actual + intended bans | All products         | Actual + intended bans |
| ATET                           |                      |                        |                      |                        |
| Treated state-waste products   | 0.128***<br>(0.0259) | 0.200***<br>(0.0287)   | 0.129***<br>(0.0349) | 0.202***<br>(0.0384)   |
| Waste type, state, and year FE | No                   | No                     | Yes                  | Yes                    |
| Controls                       | No                   | No                     | Yes                  | Yes                    |
| Observations                   | 156,949              | 48,797                 | 116,176              | 36,747                 |
| Clusters                       | 19,171               | 5,090                  | 13,552               | 3,629                  |
| Treated clusters               | 399                  | 399                    | 289                  | 289                    |
| Waste products in sample       | 179                  | 44                     | 179                  | 44                     |
| States in sample               | 169                  | 157                    | 138                  | 126                    |

Note: Fixed effects (FE) noted in table. Standard errors (SE) clustered by state-product. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Treatment = Asia-Pacific developing states (Appendix A) and China ban products (Table 1), beginning in 2017. China excluded from all models. Years covered: 1996–2020. Controls as described in text.

regarding domestic demand for raw materials: industrial output per GDP (ln) and natural resources per GDP (ln). I include state and year fixed effects. Especially important are waste type fixed effects (see again Figure 1), which avoids judgements over which product type is worse – a ranking that is non-obvious and may even be counterproductive (Carson 1962). All variables with skewed distributions and containing zeros are transformed to the log of the inverse hyperbolic sine transformation, following best practices (Aihounon and Henningsen 2021). All time-varying covariates are lagged, and standard errors are clustered by state-product.

Table 2 reports results. Altogether, there is strong support for this application of the sin tariff theory: sin tariffs have shaped the detritus of the China garbage shock. Across specifications, treated state-waste product combinations are setting tariffs from 14 per cent to 22 per cent higher than untreated combinations.

As an extension, I consider whether results in Table 2 are robust to taking into account WTO member states' bound maximum product-level tariffs. For a bound tariff to be set, a state needs to be selected into the WTO, into negotiating a binding for a specific waste product, and into the level of that binding, all of which are non-random (Pelc 2013).<sup>22</sup> Nonetheless, the expectation is that among state-waste product observations that are selected into bound maximums, treated state-waste product combinations should be more likely to minimize the gap between the tariff and the bound maximum; in jargon, this is called tariff water. I reestimate Models (3) and (4) from Table 2 with tariff water (ln) as the dependent variable, and include the negotiated bound maximum (ln) as an additional control. Table 3 reports results. As expected, the treatment has a negative and significant coefficient across specifications.<sup>23</sup> Additionally, results in Tables 2 and 3 are robust to controlling for Basel Convention membership. Results are also robust to controlling for the potentially post-treatment incidence of a national-level non-tariff measure (NTM) regulating the waste trade (Appendix 1). The interrelationship between global governance, NTMs, and tariffs in the waste trade and in environmental policy more generally is an important topic for future research.

<sup>22</sup>Within the sample, 129 Global South WTO members have set at least one bound maximum on up to 165 waste products, including 25 of the China ban waste products. Thirteen WTO members in the sample have set no bound maximums on waste products.

<sup>23</sup>Effect sizes predict decreases in tariff water of 9.8 per cent to 14.4 per cent due to the treatment, although selection issues suggest restraint in interpreting coefficient values.

**Table 3.** Less tariff water on imports treated by China garbage shock, given WTO tariff binding

|                                | (1)          | (2)                    |
|--------------------------------|--------------|------------------------|
|                                | All products | Actual + intended bans |
| ATET                           |              |                        |
| Treated state-waste products   | – 0.0935*    | – 0.135**              |
|                                | (0.0517)     | (0.0534)               |
| Waste type, state, and year FE | Yes          | Yes                    |
| Controls                       | Yes          | Yes                    |
| Observations                   | 74,992       | 23,186                 |
| Clusters                       | 7,774        | 2,163                  |
| Treated clusters               | 196          | 196                    |
| Waste products in sample       | 139          | 39                     |
| States in sample               | 107          | 105                    |

Note: Fixed effects (FE) noted in table. Standard errors (SE) clustered by state-product. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Treatment = Asia-Pacific developing states (Appendix A) and China ban products (Table 1), beginning in 2017. China excluded from all models. Years covered: 1996–2020. Controls as in Table 2, plus the bound maximum (ln). Estimations performed with *xtddidregress* in Stata 18 (Donald and Lang 2007).

### Qualitative Evidence

Quantitative analyses show robust evidence consistent with the sin tariff theory, in time series cross-sectional data and at scale. To reiterate, the Pigouvian framework establishes that these higher tariffs can generate environmental protection, even if ‘treated’ states are not motivated by the environment at all and simply increase tariffs to redistribute rents from newly competitive industries. Still, an observable implication of the sin tariff theory is that ‘treated’ actors understand that their tariff policies are making environmental–economic trade-offs. Was this the case? On the question of whether people living in ‘treated’ states understood the environmental harm resulting from the China garbage shock, myriad sources make clear that growing piles of foreign-origin garbage have been no secret.<sup>24</sup> Governments in Asia-Pacific developing states experimented with intricate domestic environmental regulations in the immediate wake of the shock; for example, at times national regulators declared that 100 per cent of containers would be inspected or that further imports would be delayed until inspection backlogs cleared.<sup>25</sup>

For evidence of taxation as an environmental policy, plastic waste diversion in particular triggered discussion throughout the region,<sup>26</sup> in states such as Indonesia,<sup>27</sup> Thailand,<sup>28</sup> Vietnam,<sup>29</sup> and Taiwan.<sup>30</sup> In 2019, the Philippines famously sent back to Canada sixty-nine containers, with then-president Duterte announcing, ‘Your garbage is on the way. Prepare a grand reception. Eat it if you want to’ (Liebman 2021). In 2022, the Philippines passed a tax of 100 pesos (USD 1.75) per kilogram of plastic waste, whether imported or domestically produced. At the same time, the Philippines has not (to date) ratified the full set of prohibitions on hazardous plastic and other

<sup>24</sup>Among many excellent journalistic and activist studies, see Franklin-Wallis (2023); Clapp (2025).

<sup>25</sup>For a detailed timeline of regulations in the region, see ‘From Green Fence to red alert: A China timeline’. Published 13 February 2018, last updated 22 March 2022. Resource Recycling. Such policies were on the whole partially implemented or walked-back, understandable given the enormous state capacity they require and the costs they would impose on the newly competitive sector.

<sup>26</sup>For a tax advisor perspective, see Turnsek, Alenka. 10 August 2021. ‘Why tax and finance functions must pay heed to plastic taxes’. EY Global.

<sup>27</sup>For a government-associated study, see for example Irianto et al. (2022).

<sup>28</sup>See for example World Bank Group (2021).

<sup>29</sup>For a scholarly treatment, see for example Retamal, Monique et al. 29 January 2019. ‘Here’s what happens to our plastic recycling when it goes offshore’. The Conversation.

<sup>30</sup>For an NGO perspective, see for example ‘Taiwan’s transition: From garbage island to recycling leader’. 18 June 2019. Rapid Transition Alliance.



waste imports available via the Basel Convention, consistent with the Pigouvian logic favoring incentive manipulation rather than outright bans.<sup>31</sup>

In Malaysia, the use of sin tariffs on plastic waste products was both explicit and well-documented. Awash in diverted imports just two months after the shock, Malaysia announced a tax of 15 ringgits (USD 3.62) per metric ton of imported waste plastic. The Malaysian Plastic Manufacturers Association complained about the tariff, calling instead to ‘protect the sustainability of the industry’ by cracking down on ‘illegal operators’. It makes sense that domestic importers would prefer a regulatory solution that preserves their cost competitiveness on international markets, with the added benefit of reducing domestic competition. At the same time, a prominent Malaysian NGO complained that ‘the cost and burden to public health and the environment far outweigh the revenue gained by the Malaysian government from the levy imposed on plastic waste imports’. It also makes sense that pro-environment interests would prefer increasing the costs of trade further. In parliament, an opposition MP asked if Malaysia would ban plastic waste imports as China had. The responsible minister demurred and said there were no such plans. In defending the government’s position, the minister cited that ‘the processing of such waste could result in RM 30 billion [USD 7.2 billion] worth of business and the government could not treat the potential lightly’.<sup>32</sup> What a sin tariff can do is trade off costs to industry against benefits to the environment, of course without fully satisfying either side.

The data suggest that Malaysia used sin tariffs on many China ban imports, at scale. Comparing trade flows from 2014 (three years pre-shock) to 2020 (three years post-shock), Malaysia’s tons of imports of China ban products grew by a factor of 4.5. Tariff revenue on China ban products grew by a factor of 3.9. That tariff revenue did not fully keep up with the volume of imports fits with the Malaysian government’s privileging of importers’ continued competitiveness. Still, Malaysian state coffers benefited. By way of comparison, Malaysian state coffers did not benefit from tariffs on other waste products: over the same period Malaysian imports of other waste products grew by a factor of 1.7, whereas tariff revenue *decreased* by a factor of 1.4. Because sin tariffs can generate environmental benefits directly, through reduced imports relative to the (unobservable) counterfactual, and/or indirectly, via the consequences of increased tariff revenues, it is difficult to discern their full environmental impact from aggregate data. Nonetheless, Malaysia’s reported direct government expenditure on environmental protection increased by a factor of 1.3 over the same 2014–20 period.<sup>33</sup>

Political economic theory makes clear that states looking to curb the adverse impact of trade have many options other than autarky. Still, the appeal of mimicking China’s outright bans has led some other states to try the same. For example, Turkey has been inundated with EU waste products as the China garbage shock caused further ripples outside the Asia-Pacific region and the study period (Gundoğdu 2024). In May 2021, Turkey announced a ban on the import of ethylene polymer waste (HS 391510), which went into effect for a week in July and was then overturned. The Turkish plastics recycling association attributed the overturn to the success of its lobbying campaign.<sup>34</sup> In 2019 India announced plans to ban all plastic waste imports. Not only did India delay for years, but when finally enacted in June 2022 enforceability was immediately in question, as the Indian plastics industry was ‘up in arms’, protesting that ‘thousands of jobs are at stake’.<sup>35</sup> Even Global South states like Turkey and India have not broken out of the sin tariff theory’s scope

<sup>31</sup>For an advocate’s perspective, see for example Ramos, Denise. 12 June 2023. ‘How did the Philippines become the world’s biggest ocean plastic polluter?’ Earth.org.

<sup>32</sup>Jun, Soo Wern. 17 November 2018. ‘Santiago: Why make Malaysia a dump for plastic waste?’ Free Malaysia Today. Staub, Colin. 16 October 2018. ‘Malaysia to enact scrap import tax and restrictions’. Resource Recycling.

<sup>33</sup>Government expenditure on environmental protection in Malaysia from 2013 to 2021, Statista, published by Amanda Siddharta (27 November 2023).

<sup>34</sup>The association estimates that the ban cost Turkey USD 547 million. ‘Polyethylene waste imports ban is lifted with active control!’ PAGEV News Release, accessed January 2023.

<sup>35</sup>Masih, Niha and Anat Gupta. ‘India imposes ban on single-use plastics. But will it be enforced?’ 1 July 2022. Washington Post.

condition, that the economic side of trade-related economic–environmental trade-offs is consequential. One implication is that those advocating for bans might remember that incremental sin tariffs can be useful environmental policies, albeit less headline-worthy.

## Conclusion

In arguing that tariffs can generate environmental protection, this article turns to the global waste trade for evidence. I introduce a new list of 179 (HS six-digit) traded waste products and track their tariffs across the Global South (1996–2020), confirming descriptive patterns consistent with the real-world use of sin tariffs. Leveraging the 2017 China garbage shock, I find quantitative support: Asia-Pacific developing state–waste product combinations ‘treated’ with simultaneous and contradictory shocks to competitiveness and environment have higher tariffs. Qualitative evidence that actors in Malaysia understood the economic–environmental trade-offs behind sin tariffs is just one example of a phenomenon playing out throughout the Global South: governments can leverage sin tariffs to influence the distribution of the physically big, environmentally overwhelming, costly leftovers of the global waste trade.

There may be more opportunities for trade policy to be repurposed in service of outcomes other than protection of material import-competing interests. Certainly, product-level tariffs have surged in popularity as much-discussed tools to achieve security and foreign policy goals. The United States–China trade war is a case in point, in which China’s targeted product-level tariff retaliation affected electoral fortunes in the United States (Kim and Margalit 2021). In the environmental realm, Shapiro (2021) argues that tariffs on carbon-intensive industries are too low, and a general upward adjustment would mitigate emissions at a low global cost, though realizing such a global tariff increase would require considerable interstate co-ordination given that greenhouse gases are inherently transboundary.<sup>36</sup> A sin tariff framework might be a way to think about EU efforts to create a carbon border adjustment mechanism (CBAM) by which taxes mitigate climate-related externalities embedded in traded goods (Zhong and Pei 2022). Tariffs might accomplish such goals, if and when the codes marking containers can be slotted into regular tariff implementation bureaucracies. For all that, it is also the case that, whatever the purpose, raising tariffs has domestic distributional effects within states. While the most productive firms find ways to continue operating despite increased costs, their less productive competitors are likely pushed out of the market (Gulotty 2020). What such unequal effects mean for subnational community and environmental outcomes is worthy of dedicated attention – especially in industries like waste processing, characterized by huge numbers of informal, marginalized, low-skilled workers (O’Neill 2019; Goldstein 2020).

Typically, studies of the politics of tariffs engage with the question of who pays the tariff. How much of this ostensible tax on foreigners is in fact passed through to importers, and eventually downstream domestic consumers? The cross-border distribution of costs is an empirical question, though the bulk of the evidence primes observers to expect domestic pass-through as the norm, not the exception. But, if you have produced trash in the Global North in recent years, you might share the intuition that exporters as well as their suppliers (such as you and me) are shouldering at least some of the costs of adjustment. Reflecting on Los Angeles’s choice in the wake of the China garbage shock to flip its policy and actually charge residents for curbside recycling services, a municipal leader lamented that once ‘the city looked at recycling as a cash inflow, and that scenario is gone’.<sup>37</sup> Just months after the China garbage shock the US Congress added ‘export incentives’ to tax reform bills, which lowered taxes on income from exports versus other sources;

<sup>36</sup>Earlier scholars have also brainstormed the net benefits of tariffs for the environment (Hecht 1997; Kraus 2000).

<sup>37</sup>Staub, Colin. 16 April 2019. ‘Shifting markets drive deeper contract negotiations’. Resource Recycling. On US city strategies, see for example ‘Beyond recycling: Policy to achieve circular waste management’, US National League of Cities Brief, 2023.

these were cheered by the chief lobbyist at the US Institute of Scrap Recycling Industries (ISRI).<sup>38</sup> In negotiations over a Global Plastics Treaty ongoing at the time of writing, the United States is pushing more trade-facilitated recycling while objecting to binding reductions in plastic production, in ways consistent with its domestic oil industry's preferences. Both Republican and Democratic administrations continue to eschew the 1989 Basel Convention that otherwise has near universal membership, retaining the ability of the United States to export hazardous waste. Such subsidies to exporters and efforts to keep global markets open reflect the reality that failure to export has outsized consequences in the waste trade.

From this point of view, sin tariffs are a means for Global South states to extract rents from the Global North in exchange for NIMBY outsourcing. If and when Global South states increase trade costs so much that export subsidies are exhausted and too many would-be waste product exports are left at home, then Global North politicians would have to devise different solutions. One might be for the Global North to ban waste-producing products; the EU's ban on single-use plastics took effect in July 2021. The problem of adjustment in the Global North has gotten some attention from scholars, policy makers, and activists, but deserves more dedicated theorizing (O'Neill 2019; Yamaguchi 2021; Wen et al. 2021; Velut and Wellhausen 2026). Here, I note only that these dynamics reinforce the political usefulness of sin tariffs in the Global South – the costs of which might be shared with or even borne by people far away.

The rubbish circumstances around the waste trade and the possibilities afforded by tariffs might provide a jumping-off point for more scholars of international relations and political economy. To be sure, the legal waste trade highlights that economic globalization, sustainability, and the normative appeal of the 'circular economy' approach come together in complex ways. Are tariffs that interfere with prices in competitive, capitalist waste markets economically optimal? Maybe not. Does the redistribution of pollution mean that protectionist states trade off short-term local benefits against long-term, aggregate global harm? Maybe so. Normative implications are not obvious. And yet, from the point of view of a government and society in a given territorial jurisdiction, encountering piles of Global North junk mail, it matters that a simple product-level tariff can alleviate consequences of low state capacity without killing off the domestic industry that transforms junk mail into something useful.

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<sup>38</sup>ISRI is the major waste and recycling trade association in the United States, which 'aggressively advocates initiatives that promote free and fair trade of recycled materials' ('ISRI: International Trade', <https://www.isri.org/advocacy-compliance/international-trade>, last accessed February 2025). Sen. Ron Johnson (R-Wisconsin) pushed for relevant tax deductions; he was previously owner and CEO of a plastics firm (Staub, Colin. 5 December 2017. 'Federal tax reform will impact plastics recycling'. Plastics Recycling Update: Resource Recycling, Inc.).

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